Name the force and its source

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify forces on a chosen object and name each interaction partner.

SUCCESS CRITERIA - I can...

1. Name the chosen object and each force source.
2. Distinguish mass from weight with units.
3. Include only forces acting on the chosen object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “interaction” and “weight” to explain your model or result. Sentence starters are optional.

interaction: two objects affecting one another

weight: the gravitational force on an object

normal force: a support force perpendicular to a contact surface

01 Bring back an earlier idea

A cart travels west at 12 m/s and west at 4 m/s 3 s later. Is it speeding up?
Find acceleration size and direction.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $F_g = mg$.
F_g is weight; m is mass. Use g = 10 N/kg for CPA. Name the source of every force.

Worked reasoning: A book rests on a table. Earth pulls the book down; the table pushes the book up. Each force label names both the force and the object producing it.

Your turn: Draw the book's force diagram.
Name the force on Earth paired with the book's weight.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Which object gets the force?

A book rests on a level table.
Draw its two vertical forces and name who exerts each.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Add and label force arrows.

The dot represents the object. Chosen object: book

Marzano check - circle: ① ② ③

04 Mass and weight

A 2 kg bag hangs at rest.
Use 10 N of weight for each kg.
Find weight. Which way does the rope pull, and how strongly?

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Name the force and its source

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 After the throw

A ball is moving up after leaving a hand. Ignore air resistance. Is the hand still pushing it?
Draw the force that remains.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: 1 2 3

06 Choose a method

A rover travels straight east at a steady 5 m/s. A second rover turns at the same steady speed. Which can have zero net force?
Explain using velocity.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: 1 2 3

07 Exit check

A 3 kg lamp hangs still from one vertical cord.
Use g=10 N/kg.
Name both forces with source, magnitude and direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A hand presses down on the resting book. Add the new force and explain how the table's support must change.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: 1 2 3